The dimension-nine threshold: an exact reduction and minimality for constructions from paired seeds

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10 October 2026 Research supplement

Abstract

The accompanying paper proves separation from the point-generated product–join class in every dimension at least nine. We show that global minimality of nine is equivalent to the unrestricted simplex bound in dimension eight, which remains unproved here. We do prove that nine is the first separation dimension within the paired two-upper-facet seed family and for all constructions from those seeds by products and joins. The proof uses a rational supporting gradient and log-concavity; a separate positive-polynomial certificate checks the seed bound. Neither argument reduces arbitrary convex bodies to these constructions. We give an equivalent section–projection inequality with no bound on the number of facets, an exact checker for individual rational polytopes, and a uniform neighbourhood bound for centred, possibly asymmetric, two-facet deformations of a simplex. Global minimality remains unproved.

1 What global minimality requires

For a full-dimensional convex body $K \subset \mathbb{R}^n$, put

$$R_n(K) = \frac{\mathrm{vol}_n(\Pi K)}{\mathrm{vol}_n(K)^{n-1}}, \qquad g(n) = \frac{n^n}{n!}, \qquad c_n = (n+1)g(n).$$

Here ΠK is the projection body. Let $\mathcal{C}$ be the class generated from a point by Cartesian products, joins, and invertible affine maps on the relevant affine hulls. The exact finite classification in [3] gives $\sup_{K \in \mathcal{C}_n} R_n(K) = c_n$ for $1 \le n \le 9$. The accompanying paper [2] supplies a strict witness in dimension nine and separation in every larger dimension. The unrestricted simplex bound in dimensions four through eight is left open in [1, Introduction].

Proposition 1. *Nine is the smallest dimension admitting separation from $\mathcal{C}$ if and only if*

$$R_8(K) \le c_8 = \frac{131072}{35} \quad \textit{for every full-dimensional convex body } K \subset \mathbb{R}^8. \tag{1}$$

Proof. The inherited pyramid calculus [4, Sections 2–5] gives, for a d-dimensional convex body K and $t \ge 0$,

$$\frac{R_{d+t}(\mathcal{P}^t K)}{c_{d+t}} = \frac{R_d(K)}{g(d)} \frac{1 + t a(K)}{d+t+1}, \qquad a(K) \ge \frac{1}{d+1}.$$

Here $a(K)$ is the lifted facet invariant of [4], extended by continuity to convex bodies. The difference between consecutive terms on the right is

$$\frac{R_d(K)}{g(d)} \frac{(d+1)a(K) - 1}{(d+t+1)(d+t+2)} \ge 0.$$

Thus any strict violation of the simplex bound in a dimension $d \le 8$ lifts to a strict violation in dimension eight. If (1) holds, no such lower-dimensional separation exists, and the known dimension-nine witness proves minimality. Conversely, a violation of (1) is already separation in dimension eight. □

By continuity of volume and the projection-body operator under Hausdorff convergence, it would suffice to prove (1) for every full-dimensional rational polytope in $\mathbb{R}^8$. This is still an unbounded class; finite searches do not establish that assertion.

2 An unrestricted section formulation

Let $\Delta_{m-1} = \{y \in \mathbb{R}^m : y_i \geq 0,\ \sum_i y_i = 1\}$ and $C_m = [-1/2, 1/2]^m$. For an eight-dimensional subspace $F \subset \mathbf{1}^\perp$ and $\lambda \in \operatorname{relint} \Delta_{m-1}$, put $S = (\lambda + F) \cap \Delta_{m-1}$. Volumes in F, and in its translates, are Euclidean eight-dimensional volumes; P_F denotes orthogonal projection.

Proposition 2. *The unrestricted bound* (1) *is equivalent to*

$$\operatorname{vol}_8(S)\, \operatorname{vol}_8(P_F C_m) \leq \frac{9}{8!} \quad \text{for every } m \geq 9,\ F, \lambda \text{ as above.} \tag{2}$$

In particular, there is no facet-count restriction in this equivalence.

Proof. Feng et al. [1, Proposition 4.1] prove that, for any such section, there is a full-dimensional polytope $Q \subset F$ with

$$R_8(Q) \geq 8^8 \operatorname{vol}_8(S) \operatorname{vol}_8(P_F C_m).$$

Thus (1) implies (2).

For the converse, let $K \subset \mathbb{R}^n$ be any full-dimensional polytope, with m facets, and translate an interior point to the origin. Let u_i be its facet-area vectors: the outward unit normal multiplied by the facet's $(n-1)$-volume. Write $z_i = h_K(u_i) > 0$ and $V = \operatorname{vol}_n(K)$. The balance and divergence identities give $\sum_i u_i = 0$ and $\sum_i z_i = nV$. Define the $m \times n$ matrix T and the offset by

$$T_i = -\frac{u_i^\mathsf{T}}{nV}, \qquad \lambda_i = \frac{z_i}{nV}, \qquad F = \operatorname{im} T. \tag{3}$$

The facet normals span $\mathbb{R}^n$, so T has rank n, $F \subset \mathbf{1}^\perp$, and λ is in the relative interior of the simplex. The facet inequalities give the exact equality $\lambda + TK = (\lambda + F) \cap \Delta_{m-1}$. Put $J = \sqrt{\det(T^\mathsf{T} T)}$. Then

$$\operatorname{vol}_n(S) = JV, \qquad \operatorname{vol}_n(P_F C_m) = \frac{1}{J} \sum_{|I|=n} |\det T_I|.$$

For the second identity, $U = T(T^\mathsf{T} T)^{-1/2}$ has orthonormal columns, and the zonotope determinant formula gives the sum of the absolute minors of U. Cauchy's projection formula similarly gives $\operatorname{vol}_n(\Pi K) = \sum_{|I|=n} |\det(u_i)_{i \in I}|$. Consequently

$$\operatorname{vol}_n(S) \operatorname{vol}_n(P_F C_m) = V \sum_{|I|=n} |\det T_I| = \frac{R_n(K)}{n^n}. \tag{4}$$

Applying (2) with $n = 8$ proves the simplex bound for every eight-dimensional polytope. Hausdorff approximation then proves it for every convex body. □

The section construction and one direction are prior work in [1]; the exact representation above spells out the converse relevant to the unrestricted goal. A proof for $m = 10$, or for central hyperplane sections alone, would not establish (2). Recent sharp slicing and stability results [6, 7] concern section volume rather than this coupled section–projection product. For example, in the central $m = 10$ case, combining Webb's density bound $p_a(0) \leq 1/\sqrt{2}$ with $D(a) := \sum_{i<j} |a_i - a_j| \leq \sqrt{330}$ for $|a| = 1$ gives only $D(a) f_a(0) \leq 9\sqrt{165} > 81$, since $f_a(0) = 9 p_a(0)$. The required coupled bound is $D(a) f_a(t) \leq 81$ by [1, Proposition 4.2], for every admissible offset t. Thus those separate sharp estimates do not by themselves close even this special case, and no reduction to bounded m is asserted here.

3 A proved minimum for the paired family

For $0 \le u_1 \le \cdots \le u_m < 1$ and $n \ge 2m$, define

$$K_n(u) = \{x \in \mathbb{R}^n : x_i \ge 0,\ \mathbf{1}\cdot x + |b \cdot x| \le 1\}, \qquad b = (-u_m, \dots, -u_1, u_1, \dots, u_m, 0^{n-2m}).$$

For $m = 0$ this means the standard simplex. The family is inherited from the two-facet constructions of [1]; we make no priority claim for the family or the dimension-nine phenomenon.

Theorem 3. *Every paired seed $K_n(u)$ in dimension $n \le 8$ satisfies $R_n(K_n(u)) \le c_n$. The paired seed $K_9(1/100, 2/100, 3/100, 5/100)$ satisfies*

$$\frac{R_9}{c_9} = \frac{20811283583}{20797677600} > 1.$$

Consequently nine is the first separation dimension within this paired seed family, with zero coordinates and affine images allowed.

We first record an elementary consequence of log-concavity.

Lemma 4. *If $\mathbb{P}(Y > 0) = 1$, Y has a log-concave survival function, and its mean M is finite, then $\mathbb{E}e^{-Y} \le (1+M)^{-1}$.*

Proof. Write $S(t) = \mathbb{P}(Y > t)$, so $S(0) = 1$. Convexity of $-\log S$ implies $S(t+s) \le S(t)S(s)$ for $s, t \ge 0$. Consequently

$$J(t) := \int_t^\infty S(s)\, ds \le MS(t).$$

For $L = \int_0^\infty e^{-t}S(t)\, dt$, Tonelli's theorem gives $\int_0^\infty e^{-t}J(t)\, dt = M - L \le ML$. Hence $L \ge M/(1+M)$ and $\mathbb{E}e^{-Y} = 1 - L \le (1+M)^{-1}$. □

Proof of Theorem 3. It is enough to consider dimension eight: appending a zero tilt is a pyramid, so Proposition 1's monotonicity transfers the eight-dimensional bound to smaller dimensions. Padding with zero pairs then gives four parameters $0 \le u_1 \le \cdots \le u_4 < 1$. Let X_i be independent standard Laplace variables, each with density $e^{-|x|}/2$, and write

$$M(u) = \mathbb{E}\left|\sum_{i=1}^4 u_i X_i\right|, \qquad I(u) = 8!\,\mathrm{vol}_8(K_8(u)).$$

The homogeneous gauge integral gives

$$I(u) = \int_{\mathbb{R}^8_+} e^{-\mathbf{1}\cdot x - |b\cdot x|}\, dx = \mathbb{E}e^{-Y}, \qquad Y = \left|\sum_{i=1}^4 u_i X_i\right|,$$

because the difference of two independent standard exponential variables is standard Laplace. If $u \ne 0$, the sum has a symmetric log-concave density by Prékopa's convolution theorem [5, Theorem 7]. The density of Y on $[0,\infty)$ and its survival function are therefore log-concave; the latter follows by integrating $f(t+s)$ over $s \ge 0$ using [5, Theorem 6]. Lemma 4 yields $I(u) \le (1+M(u))^{-1}$.

The convex homogeneous function M on $\mathbb{R}^4$ has, at $v = (1,2,4,8)$, supporting gradient

$$\nabla M(v) = \frac{1}{18225}(2749, 5304, 9624, 15664). \tag{5}$$

For a direct exact calculation, partial fractions of the characteristic function $\prod_i (1+u_i^2t^2)^{-1}$ give

$$M(u) = \sum_{i=1}^4 \frac{u_i^7}{\prod_{j\ne i}(u_i^2 - u_j^2)} \quad (0 < u_1 < \cdots < u_4).$$

Differentiating at v gives (5). Convexity and homogeneity imply $M(u) \geq \nabla M(v) \cdot u$. Writing $x_1 = u_1$ and $x_j = u_j - u_{j-1}$ for $j \geq 2$, we obtain

$$M(u) - \frac{u_1 + 3u_2 + 5u_3 + 7u_4}{9} \geq \frac{941x_1 + 217x_2 + 988x_3 + 1489x_4}{18225} \geq 0. \tag{6}$$

The seed formula in [2, Proposition 2] now gives, with $w \cdot u = u_1 + 3u_2 + 5u_3 + 7u_4$,

$$\frac{R_8(K_8(u))}{c_8} = I(u)\left(1 + \frac{w \cdot u}{9}\right) \leq \frac{1 + w \cdot u/9}{1 + M(u)} \leq 1.$$

For $u = 0$ the body is a simplex; repeated and zero parameters also follow by continuity, or directly from the integral argument. The displayed rational value in dimension nine is checked by the exact seed formula and the independent geometric certificate in [2]. Since the class maximum there is c_9, this is separation from $\mathcal{C}_9$. □

3.1 Closure under arbitrary products and joins

Let $\mathcal{F}$ be the class generated by a point and all paired seeds $K_n(u)$, using Cartesian products, joins, and invertible affine maps. This allows arbitrary finite operation trees, rather than only pyramids or the balanced recursion of the accompanying paper.

Proposition 5. *For $1 \leq n \leq 8$ and every $K \in \mathcal{F}_n$, one has $R_n(K) \leq c_n$. Thus nine is the first separation dimension within $\mathcal{F}$.*

Proof. Use the inherited coordinates $H = 1/a$ and $Q = aR/g(n)$ from [4]. The universal invariant bounds give $1 \leq H \leq n + 1$. We prove by induction on operation trees of output dimension at most eight that

$$Q(H + 8 - n) \leq 9. \tag{7}$$

The formal point state is $n = 0$, $H = Q = 1$, which satisfies the bound. For a seed of dimension $n \leq 8$, the left side divided by nine is $R_8(\mathcal{P}^{8-n}K)/c_8$. These pyramids remain paired seeds, so Theorem 3 proves the base cases.

For a join of factors with dimensions r, s, states $(h, q), (j, v)$, and $n = r + s + 1 \leq 8$, the new state is $(h + j, qv)$. Put $A = h - r - 1 \leq 0$ and $B = j - s - 1 \leq 0$. Then

$$(h + 8 - r)(j + 8 - s) = (9 + A)(9 + B) \geq 9(9 + A + B) = 9(h + j + 8 - n).$$

The two input bounds therefore imply the output bound.

For a product, assume $1 \leq r \leq s$ and $n = r + s \leq 8$; products with a point do not change a state. With $G = g(r)g(s)/g(n)$, the operation rules give

$$H = \frac{nhj}{rj + sh}, \qquad Q = qvG\frac{rj + sh}{n}.$$

It suffices to verify

$$(h + 8 - r)(j + 8 - s) - 9G\left(hj + \frac{8 - n}{n}(sh + rj)\right) \geq 0. \tag{8}$$

This is bilinear in (h, j), so its minimum on $[1, r + 1] \times [1, s + 1]$ occurs at a corner. For the sixteen pairs $1 \leq r \leq s$, $r + s \leq 8$, all 64 corner values are at least nine; the checker recomputes each exact value. This proves product closure.

Finally, for $n \leq 8$, inequality (7) implies

$$\frac{R_n(K)}{c_n} = \frac{HQ}{n + 1} \leq \frac{9H}{(n + 1)(H + 8 - n)} \leq 1,$$

where the last inequality is $(8 - n)H \leq (8 - n)(n + 1)$. The known paired witness in dimension nine belongs to $\mathcal{F}$ and exceeds the class maximum c_9. □

4 Centred asymmetric cuts near a simplex

The supporting-gradient calculation also gives a local result without paired coefficients. Let $E_1, \dots, E_8$ be independent standard exponential variables, and define

$$M(b) = \mathbb{E}\left|\sum_{i=1}^{8} b_i E_i\right|, \qquad D(b) = \sum_{i<j} |b_i - b_j|, \qquad r(b) = \max_i b_i - \min_i b_i.$$

Lemma 6. *For every $b \in \mathbb{R}^8$,*

$$M(b) \geq \frac{D(b)}{18} + \frac{217}{36450} r(b). \tag{9}$$

Proof. The function M is convex, positively homogeneous, and invariant under coordinate permutations. At $v = (-8, -4, -2, -1, 1, 2, 4, 8)$ its gradient is

$$\nabla M(v) = \frac{1}{36450}(-15664, -9624, -5304, -2749, 2749, 5304, 9624, 15664).$$

This is half the paired gradient, with the appropriate signs: swapping every positive and negative coordinate changes the sign of $\sum v_i E_i$, so the two component derivatives in a pair are opposite. It can also be differentiated independently from the exponential divided-difference formula, as the checker does. Homogeneity and the supporting inequality give $M(b) \geq \nabla M(v) \cdot b$. Assume $b_1 \leq \cdots \leq b_8$. Then $D(b) = \sum_i (2i - 9) b_i$. The coefficient difference between $\nabla M(v) \cdot b$ and $D(b)/18$ sums to zero and has the seven tail sums

$$\frac{1}{36450}(1489, 988, 217, 941, 217, 988, 1489).$$

Writing b in its seven nonnegative successive gaps proves the claimed bound, because their sum is $r(b)$. □

Proposition 7. *Let $b \neq 0$, $\sum_i b_i = 0$, and*

$$K_\varepsilon(b) = \{x \in \mathbb{R}^8 : x_i \geq 0,\ \mathbf{1} \cdot x + \varepsilon |b \cdot x| \leq 1\}.$$

At $\varepsilon = 0$, the right derivative of $R_8(K_\varepsilon(b))/c_8$ is at most $-217 r(b)/36450 < 0$. More explicitly, for $0 < s := \varepsilon r(b) \leq 1/500$,

$$\frac{R_8(K_\varepsilon(b))}{c_8} \leq 1 - \frac{s}{3000} < 1. \tag{10}$$

The coefficients need not occur in opposite pairs.

Proof. Write $Z = \sum_i b_i E_i$, $p = \mathbb{P}(Z > 0)$ and $I(\varepsilon) = \mathbb{E} e^{-\varepsilon |Z|} = 8! \operatorname{vol}_8(K_\varepsilon(b))$. Put

$$\alpha_\pm = \frac{\mathbb{E}(e^{-\varepsilon |Z|} \mathbf{1}_{\{\pm Z > 0\}})}{I(\varepsilon)}, \quad t = \alpha_+ - \alpha_-, \quad d_i = 1 + \varepsilon t b_i.$$

Integrating the homogeneous gauge on each of the two cones shows that $\alpha_\pm$ are their fractions of the total origin-pyramid volume. The facet-area vectors divided by $8 \operatorname{vol}_8(K_\varepsilon(b))$ are $-d_i e_i$ and $\alpha_\pm(\mathbf{1} \pm \varepsilon b)$. Enumerating the minors with zero, one, or two upper-facet vectors gives

$$\frac{R_8(K_\varepsilon(b))}{c_8} = I(\varepsilon) \left[\prod_i d_i + \frac{2\varepsilon \alpha_+ \alpha_-}{9} \sum_{i<j} |b_i - b_j| \prod_{k \neq i,j} d_k\right]. \tag{11}$$

Differentiation at zero, using $\sum b_i = 0$, yields

$$\frac{d}{d\varepsilon}\frac{R_8}{c_8}\Big|_{0+} = -M(b) + \frac{2p(1-p)}{9}D(b) \le -M(b) + \frac{D(b)}{18}.$$

Lemma 6 proves the derivative assertion.

For the finite bound, centring gives $|b_i| \le r := r(b)$ and $\mathbb{E}Z^2 = \sum_i b_i^2 \le 2r^2$. The latter follows by summing $(b_i - \min b)(b_i - \max b) \le 0$ and completing the square. Since $e^{-y} \le 1 - y + y^2/2$ for $y \ge 0$, $I(\varepsilon) \le 1 - \varepsilon M(b) + s^2$, as well as $I \le 1$. All $d_i \ge 1 - s > 0$, and $\sum_i (d_i - 1) = 0$ implies $\prod_i d_i \le 1$. Hence (11), $\alpha_+\alpha_- \le 1/4$, and Lemma 6 give

$$\frac{R_8}{c_8} \le 1 - \frac{217}{36450}s + s^2 + \frac{\varepsilon D(b)}{18}((1-s)^{-2} - 1).$$

The ordered-gap expression for D gives $D(b) \le 16r$. For $0 \le s \le 1/500$, elementary rational bounds give

$$(1-s)^{-2} - 1 \le \frac{201}{100}s, \qquad \frac{217}{36450} - \frac{209}{75}\frac{1}{500} \ge \frac{1}{3000}.$$

Substitution proves (10). The checker verifies these rational constants separately from the analytic argument. □

This proposition excludes a uniform neighbourhood of the simplex in the centred two-facet family. It is not a bound for all asymmetric cuts: dropping $\sum_i b_i = 0$ adds $(2p-1)\sum_i b_i$ to the first derivative. Nor does a negative derivative at a simplex control arbitrary bodies farther away.

5 Independent arithmetic check and remaining scope

The file `check_minimality.py` recomputes the supporting gradient, the four margins in (6), the dimension-nine rational ratio, and every corner obligation in (8). An unpruned enumeration of all operation states gives the simplex maximum in dimensions one through nine, independently cross-checking the previously published Pareto classification [3].

There is also a separate algebraic proof of the eight-dimensional family bound. For distinct positive parameters, write

$$I(u) = \frac{N(u)}{D(u)}, \qquad D(u) = \prod_i (1 + u_i) \prod_{i<j} (u_i + u_j).$$

The supplied integer numerator N has 125 monomials. The checker reconstructs D and verifies the divided-difference formula for I after multiplication by the Vandermonde polynomial. It then substitutes $u_i = x_1 + \cdots + x_i$ into

$$9D(u) - (9 + w \cdot u)N(u).$$

The resulting polynomial has 585 nonzero monomials, all with positive integer coefficients, of total degrees seven through ten. This proves the bound throughout the ordered positive parameter region; continuity handles its boundary. The checker uses standard-library integer and rational arithmetic, with no computer-algebra system or optimizer.

Run the following commands from this supplement's directory:

```
python -B check_minimality.py --check-report
python -B -m unittest -v test_minimality.py
python -B -O -m unittest -v test_minimality.py
```

The tests reject altered numerators and denominators, false or missing gap coefficients, duplicate monomials, and invalid exponents. The analytic arguments are supplied above, rather than inferred from finite checks, and have not been proof-assistant formalized or externally peer reviewed.

The separate file `check_unrestricted.py` accepts rational halfspaces, a finite set of feasible points, a positive oriented full-dimensional simplicial chain, and a generic interior probe. There is no paired-coefficient or upper facet-count condition. Interior faces must cancel; every remaining face must lie on a unique input facet; and the probe must have multiplicity one. The integer multiplicity is constant away from the chain's faces. Crossing an interior face does not change it because the oriented boundary cancels, and the interior of a convex polyhedron is connected. Thus these conditions prove coverage with multiplicity one; they also exclude an unbounded input polyhedron, since a finite chain is compact. This is the positive-chain method already used by the original paper's seed checker, now applied to general facet data.

The checker reconstructs facet-area vectors from boundary minors, checks balance and the divergence identity, constructs (3), and verifies (4). It records rational squared Euclidean volumes, so no rounding of square roots enters acceptance. Six supplied controls include the known eight- and nine-dimensional seeds and an eight-dimensional polytope with three upper facets and eleven facets in total. A centred asymmetric control has tilt range exactly $1/500$ and verifies the finite local bound. None is an unrestricted counterexample in dimension eight. Eight additional tests check positive controls and reject altered geometry and chains.

```
python -B check_unrestricted.py --check-report
python -B -m unittest -v test_unrestricted.py
python -B -O -m unittest -v test_unrestricted.py
```

Checking individual inputs cannot establish the universal quantifier in (2). An unrestricted proof, or a strict rational counterexample passing the geometric checks, is still required.

Theorem 3 and Proposition 5 exclude the paired family and its complete product–join closure as sources of a lower-dimensional witness. General asymmetric seeds and arbitrary convex bodies need not belong to this generated class. The results do not prove (1), nor imply that every maximizer belongs to $\mathcal{F}$. Global minimality is therefore open in this work. The next substantive requirement is a bound covering all eight-dimensional convex bodies, equivalently (2) for unbounded m, or an exact counterexample refuting it. Proposition 7 and the general geometry interface provide additional tools for that task; they do not replace it.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

Data and code availability

This note, its source, polynomial certificate, checker, and exact-arithmetic report are supplied in the accompanying reproduction repository [2], under `research/minimality/`. The original paper and its versioned release remain separate from this research supplement. The supplied checks establish the stated arithmetic obligations; they do not claim an unrestricted nonexistence proof or global bibliographic priority.

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